Phase-Locked Loops

Phase-Locked Loops

A Phase-Locked Loop (PLL) is a **negative feedback control system** that automatically synchronizes the phase and frequency of an internal oscillator with a reference input signal. It **generates a stable output frequency** related to the input by tracking and minimizing any phase error.

Applications of Phase-Locked Loop

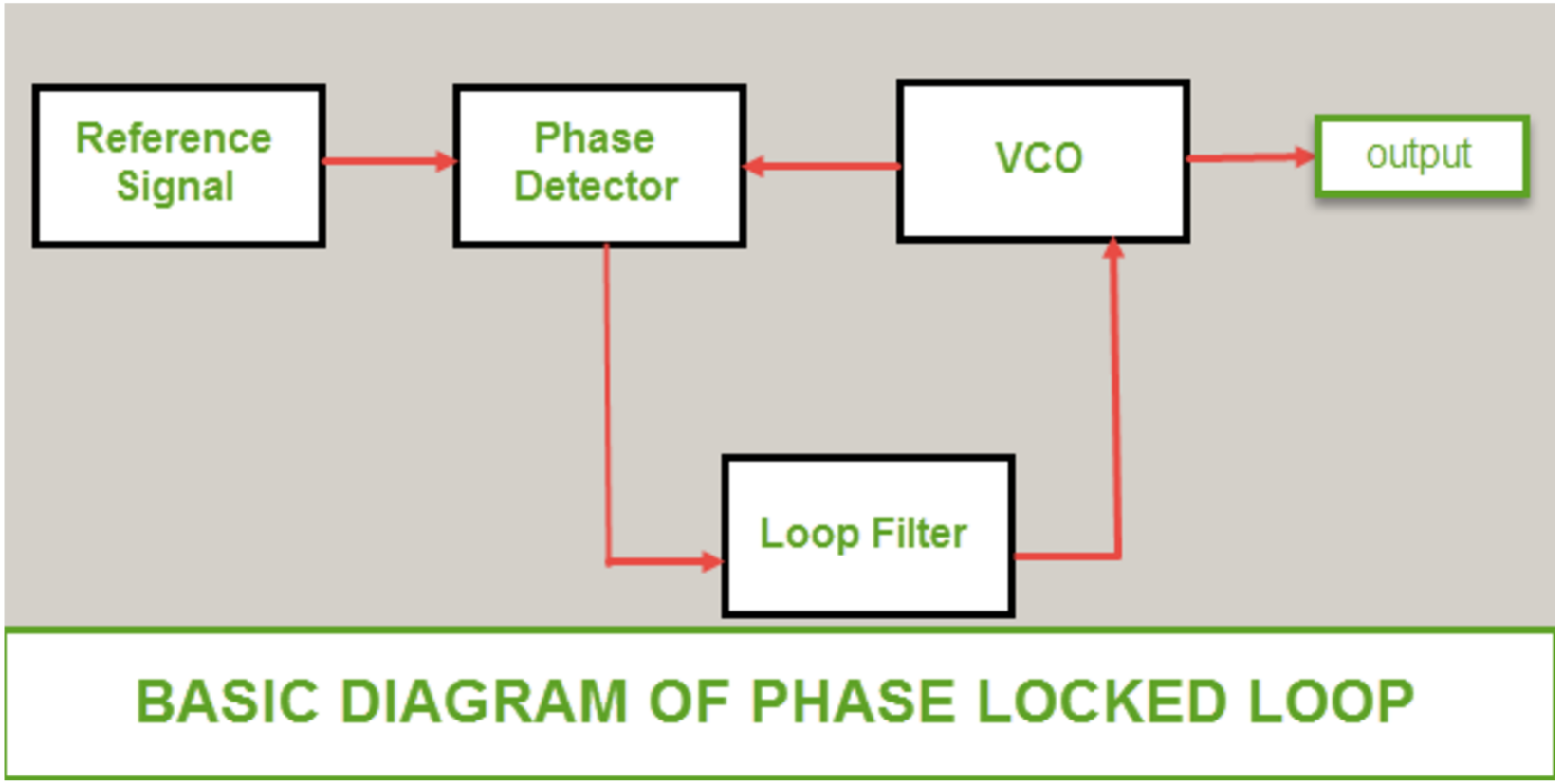
Phase-locked loop (PLL) circuits exist in a wide variety of high frequency applications, from simple clock clean-up circuits, to local oscillators (LOs) for high performance radio communication links, and ultrafast switching frequency synthesizers in vector network analyzers (VNA).

PLLs are used to generate, stabilize, modulate, demodulate, filter or recover a signal from a "noisy" communications channel where data has been interrupted.

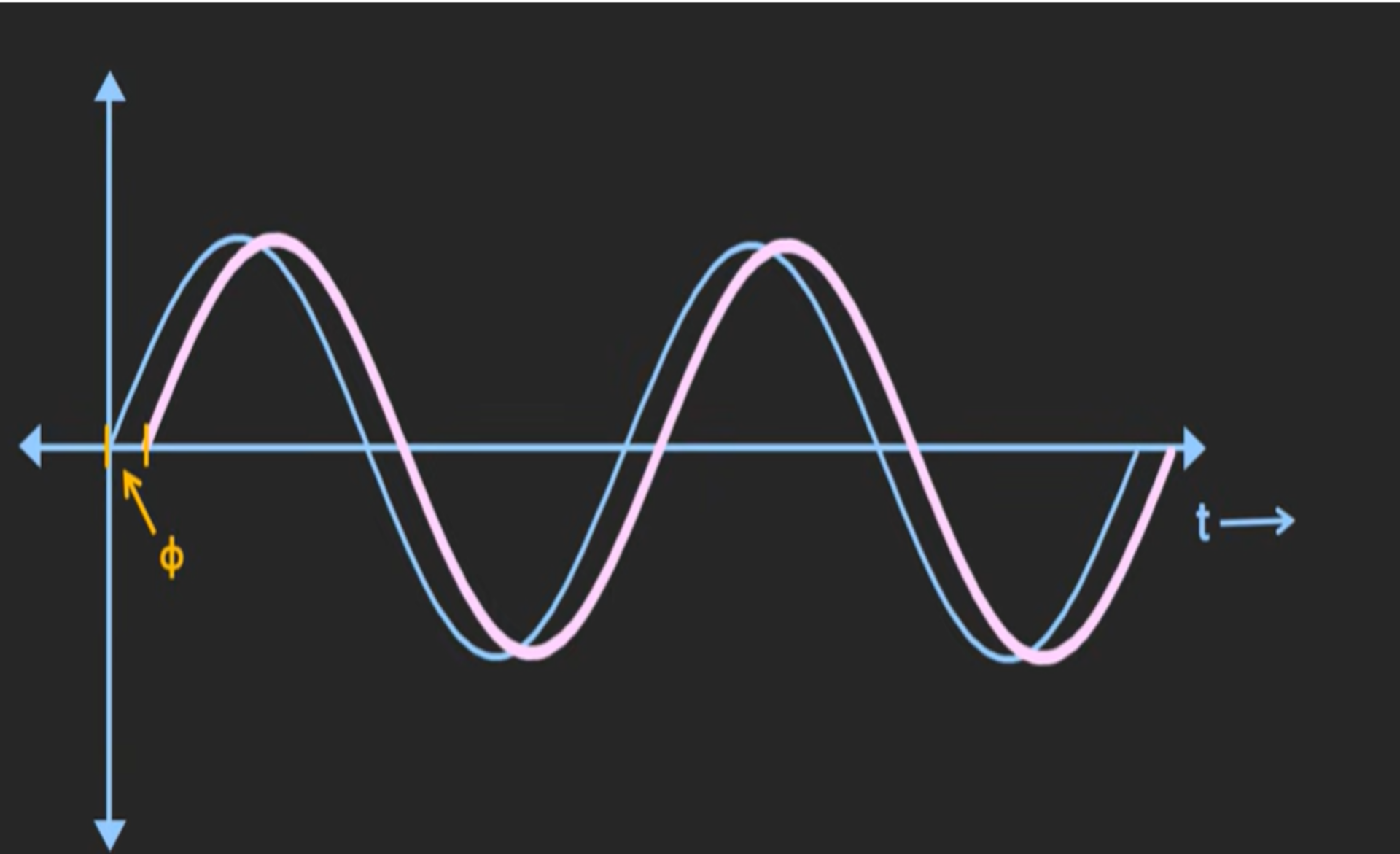
Applications of Phase-Locked Loop

- FM demodulation networks for FM operations
- It is used in motor speed controls and tracking filters.
- It is used in frequency shifting decodes for demodulation carrier frequencies.
- It is used in time to digital converters.
- It is used for Jitter reduction, skew suppression, clock recovery.

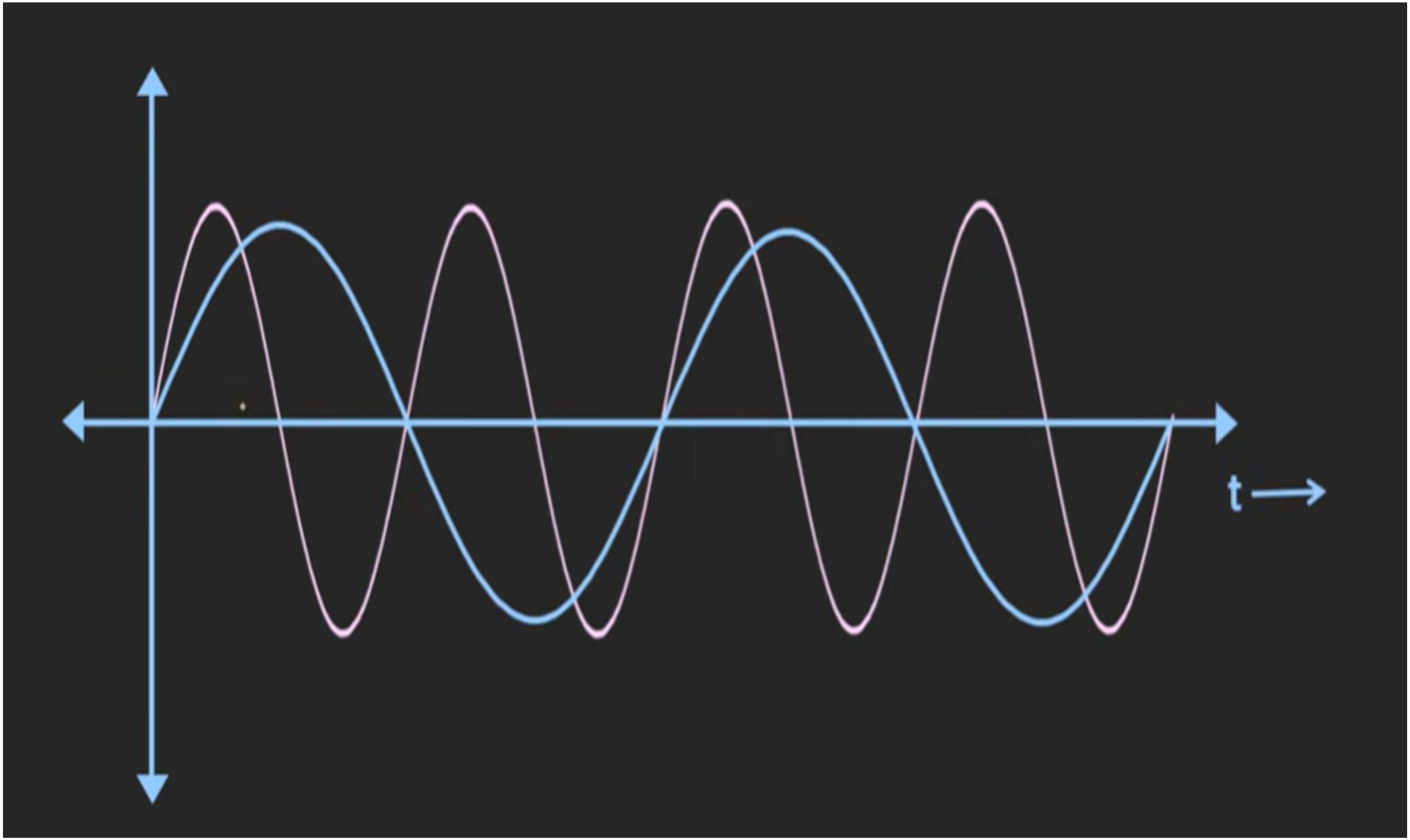
Phase-Locked Loops



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Phase-Locked Loop

- **Phase/Frequency Detector (PFD) :** Compares the phase and frequency of the reference signal against the feedback signal.
- Generates an error signal (voltage or current) proportional to the phase difference between the two signals

Phase-Locked Loop

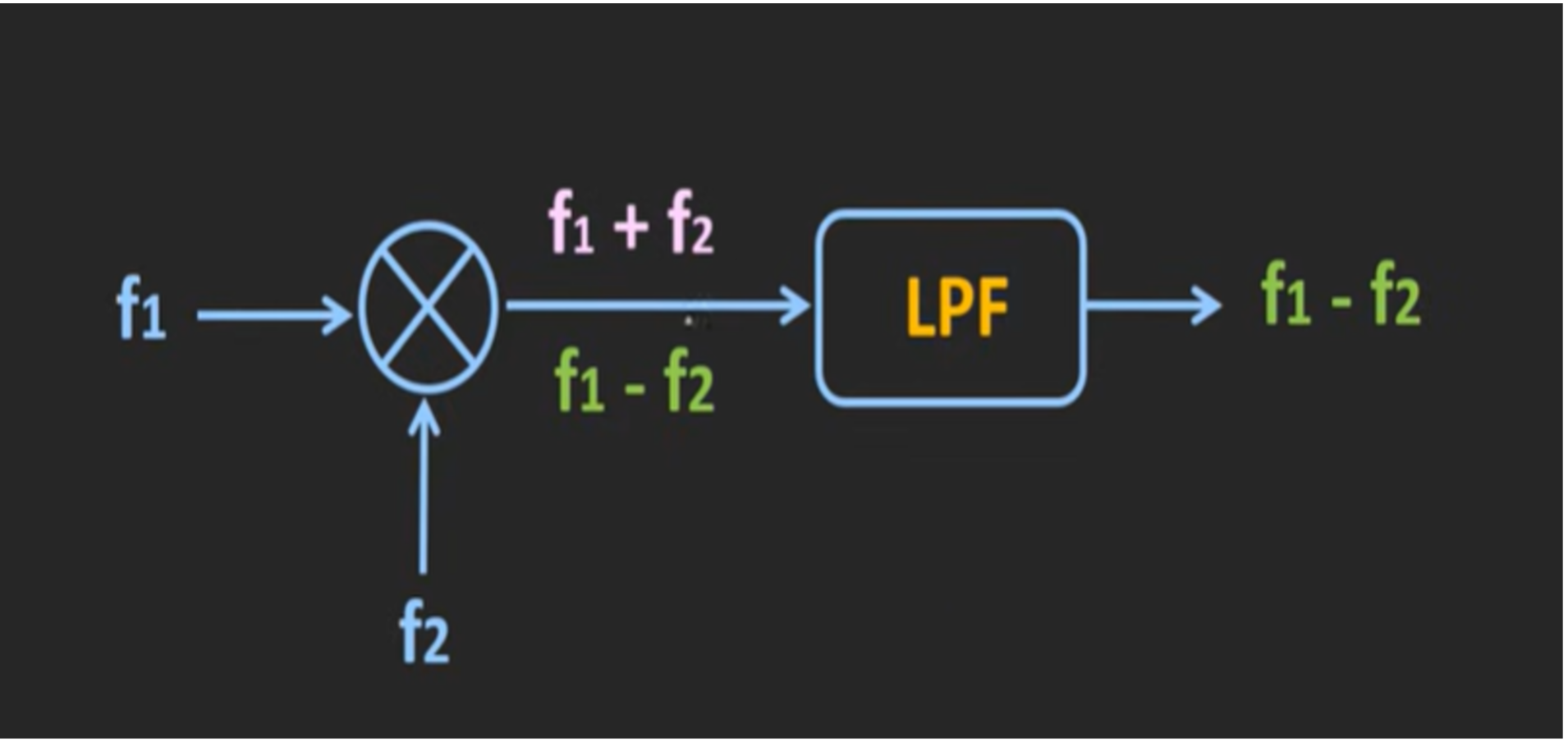
Phase Detector

$\alpha \quad \beta \Rightarrow$ Phase Difference $= \alpha - \beta \approx \sin(\alpha - \beta)$

$$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \sin\beta\cos\alpha$$

$$\sin\alpha\cos\beta = \frac{\sin(\alpha - \beta)}{2} + \frac{\sin(\alpha + \beta)}{2}$$

Phase-Locked Loop



Phase-Locked Loop

- **Loop Filter:** A low-pass filter that smooths the detector's output
- Filters out high-frequency noise, spurs, and AC components from the phase detector's output.
- Converts the error signal into a smooth, steady DC control voltage

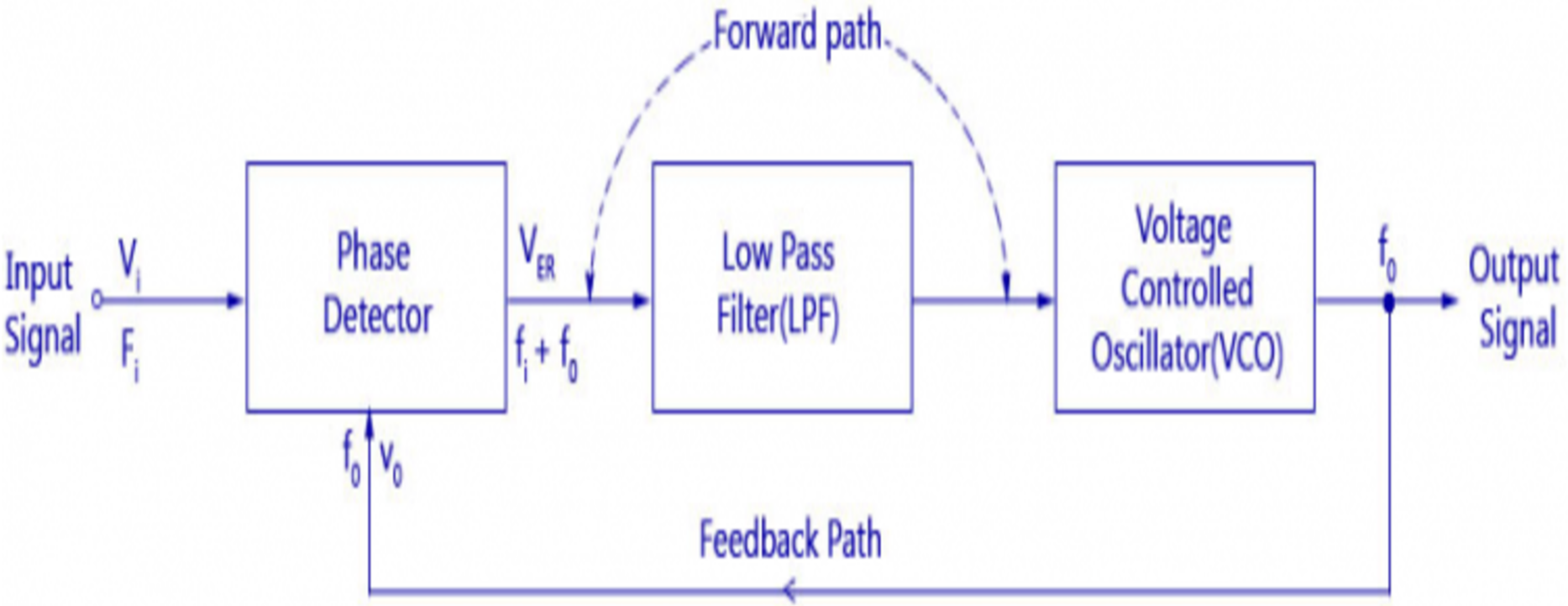
Phase-Locked Loop

- **Voltage-Controlled Oscillator (VCO) :** Adjusts its output oscillation frequency proportionally to the input control voltage.
- Generates an output frequen that changes linearly with the applied DC control voltage from the LPF.
- If the control voltage shifts, the VCO speeds up or slows down its oscillation frequency to match the reference.

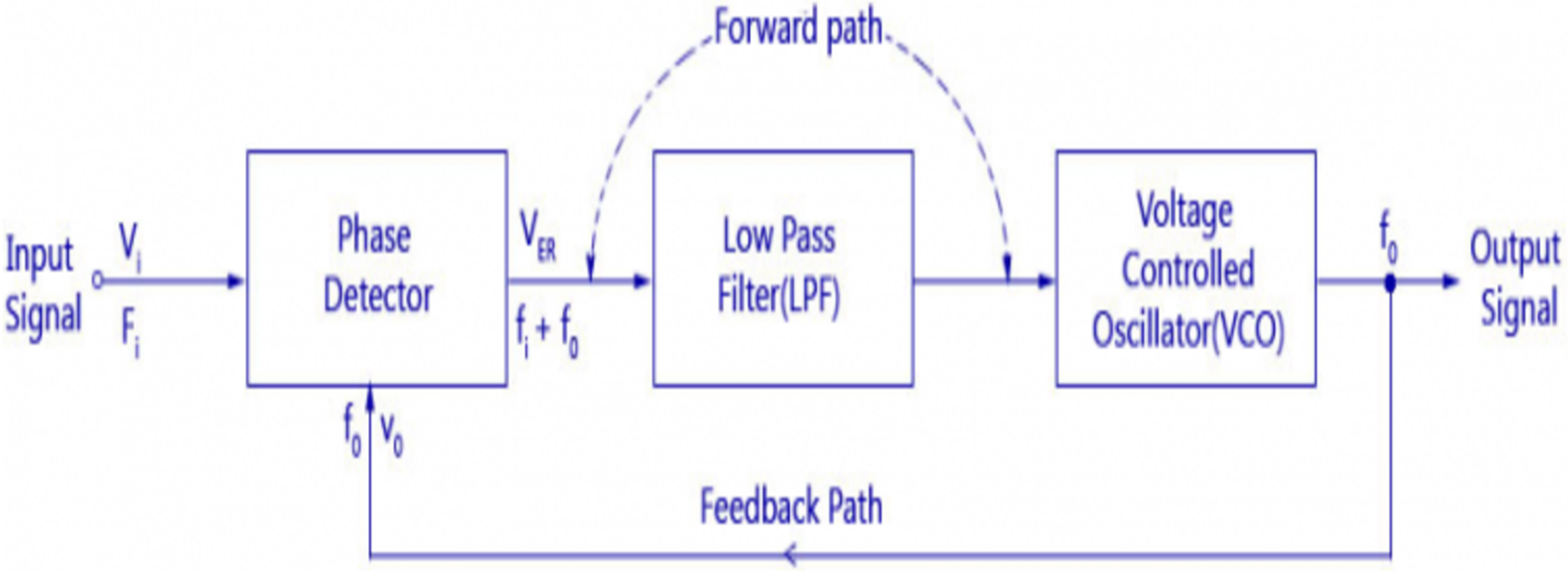
Phase-Locked Loop

- **Feedback Divider (N):** A frequency divider placed in the feedback loop that allows the output frequency fo to be a whole-number or fractional multiple of the reference frequency f_{ref}, modeled as $fo = N * f_{ref.}$

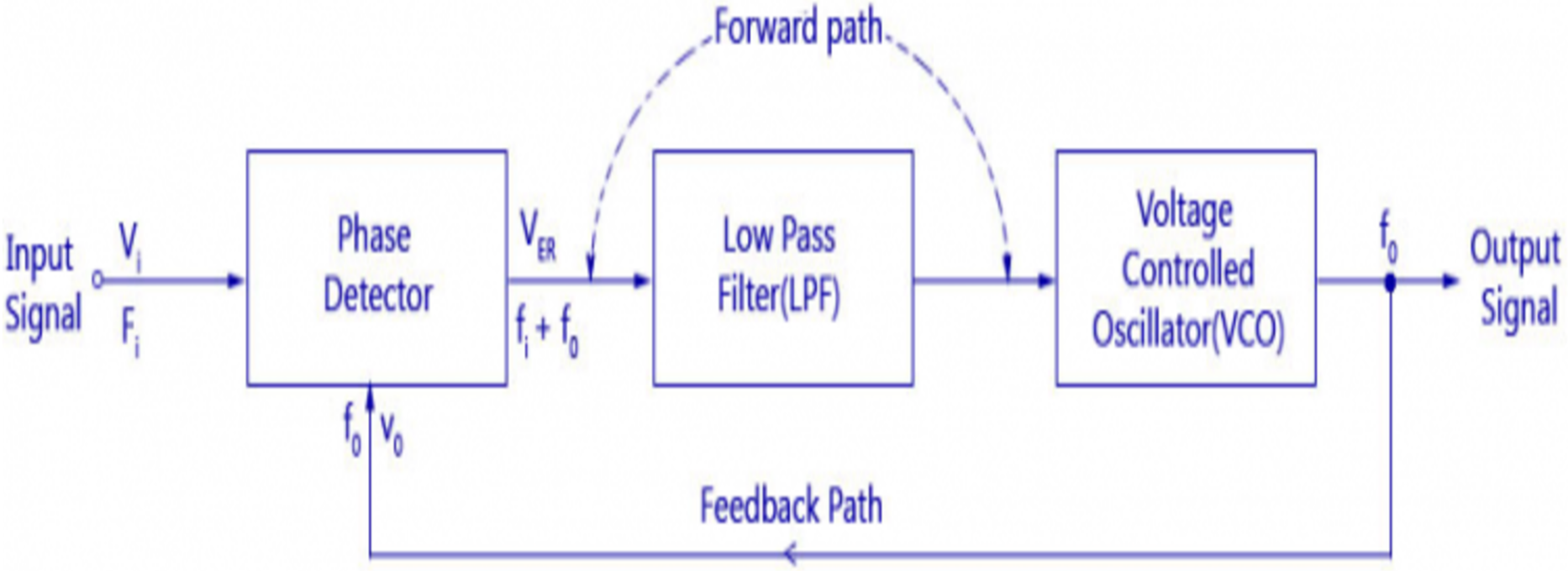
- Allows the output frequency to be a clean, locked integer or fractional multiple of the reference input, turning the PLL into a **frequency synthesizer**



The phase-locked loop consists of a phase detector, a voltage controlled oscillator and, in between them, a low pass filter is fixed.



The input signal ‘Vi’ with an input frequency ‘Fi’ is conceded by a phase detector. Basically the phase detector is a comparator that compares the input frequency fi through the feedback frequency fo.



The output of the phase detector is (fi+fo) which is a DC voltage. The out of the phase detector, i.e., DC voltage is input to the low pass filter (LPF); it removes the high-frequency noise and produces a steady DC level, i.e., Fi-Fo.

Phase-Locked Loops: States of Operation

- **Free-Running:** The loop is open or out of range; the VCO operates at its intrinsic natural frequency.
- **Capture (Pull-In):** The loop detects the reference signal and begins shifting the VCO frequency closer to the input frequency.
- **Locked:** The phase and frequency difference between the reference and feedback signals remains constant (zero frequency error), maintaining continuous synchronization.

Phase-Locked Loops

A PLL reduces phase errors between output and input frequencies.

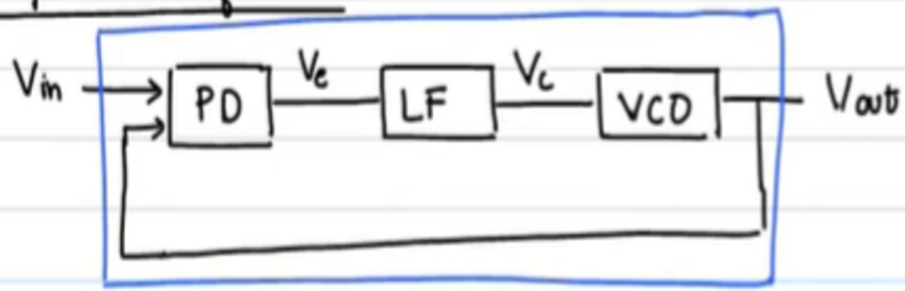
When the phase difference between these signals is zero, the system is said to be "locked." And this locking action depends on the PLL's ability to provide negative feedback -- i.e., route the output signal back to the phase detector.

Phase-Locked Loops

- It generates an error signal that corresponds to the phase difference between the signals. This difference is passed on to the low-pass filter that removes any high-frequency elements, and filters the error signal into a varying DC level. This "feedback signal" is then applied back to the voltage-controlled oscillator to control its frequency.

Phase-Locked Loops

Basic Operation of PLL



Block diagram of a PLL

PD : Phase Error Detector
LF : Loop Filter
VCO : Voltage Controlled Oscillator

$V_{in} = A_{in} \sin(\omega_{in} \cdot t)$; $\phi_{in}(t) = \omega_{in} \cdot t$

$V_{out} = A_{out} \sin(\omega_{out} \cdot t)$; $\phi_{out}(t) = \omega_{out} \cdot t$

$\phi_{er}(t) = \phi_{in}(t) - \phi_{out}(t)$

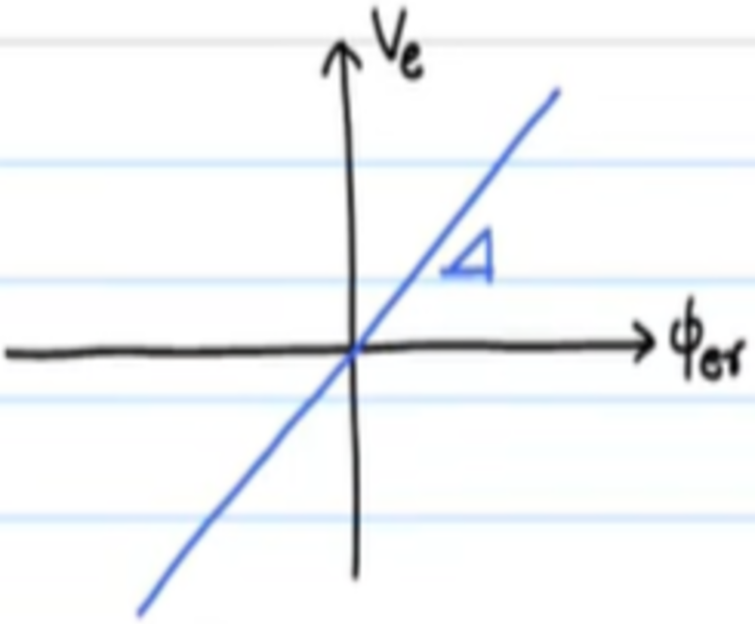
PD output, $V_e \propto \phi_{er}$

Phase-Locked Loops

PD o/p. $V_e \propto \phi_{er}$

Gain of PD, $K_{PD} = \frac{dV_e}{d\phi_{er}}$

$[K_{PD}] = V/rad$



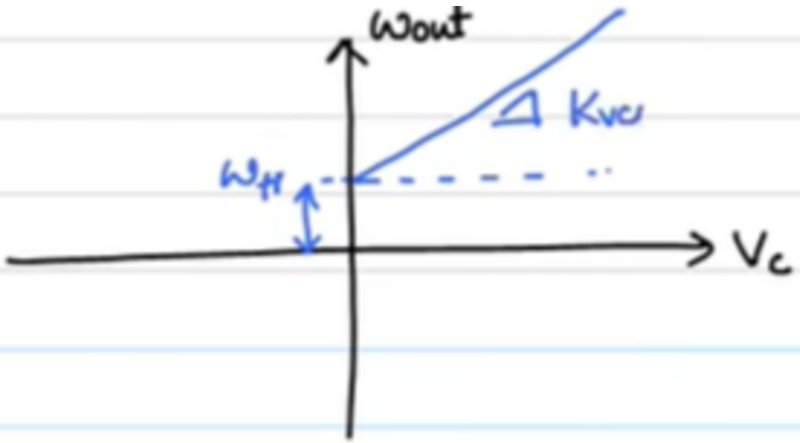
Phase-Locked Loops

Loop Filter Transfer Function, $LF(s) = \frac{V_c(s)}{V_e(s)}$

$\omega_{out} = \omega_{fr} + K_{vco} \cdot V_c$

ω_{fr}: free running frequency

K_{vco}: gain of the oscillator



- $\omega_{in} = \omega_{out}$ in steady state / locked state
- $\frac{d\phi_{er}}{dt} = 0$ in steady state /

Phase-Locked Loops

1. Lock Range (ΔfL)

The lock range is the maximum frequency band over which the loop can track input frequency changes continuously without losing synchronization.

The Mechanism: If the input frequency slowly shifts away from the center frequency (f_o), the phase detector generates an increasing control voltage to force the VCO to follow it.

The Limit: The boundary of the lock range is reached when the Phase Detector hits its maximum possible output voltage or the VCO hits its maximum tuning limit.

Phase-Locked Loops

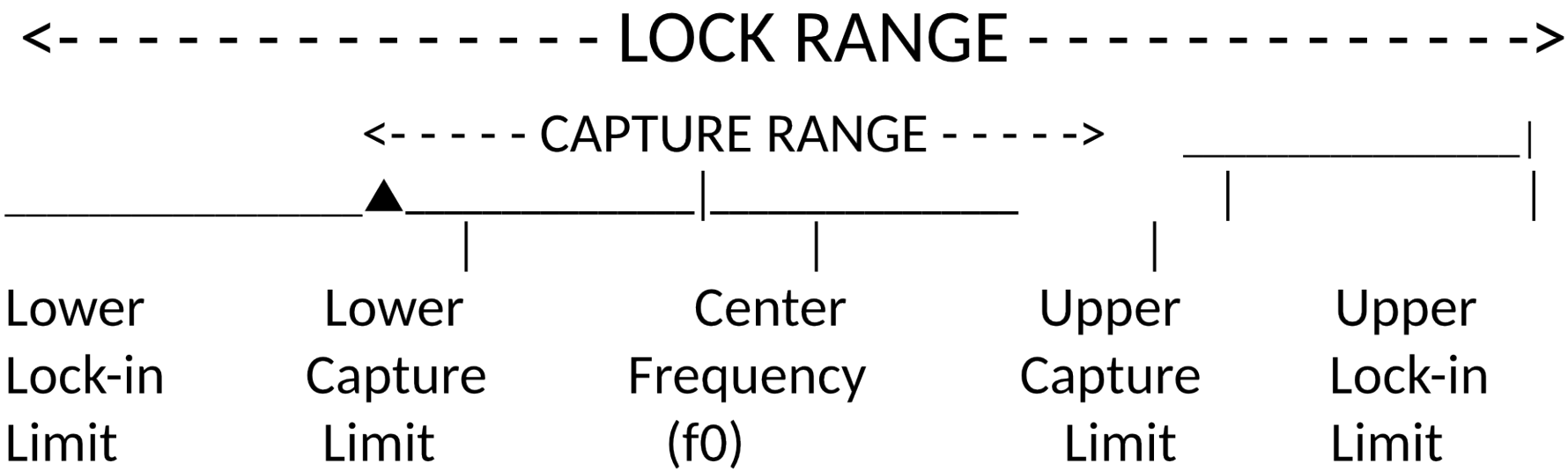
2. Capture Range (Δ fC)

The capture range is the frequency band centered around the VCO's free-running frequency (f_o) where the PLL can successfully pull itself into lock.

The Mechanism: When the loop is unlocked, the Phase Detector outputs a high-frequency "beat note" equal to the difference between the input and VCO frequencies (fin - f0).

The Limit: For capture to happen, this beat note must be low enough to pass through the Low-Pass Filter. If the filter cuts off the beat note completely, the VCO receives no control voltage shift, and the loop cannot lock.

Phase-Locked Loops



Phase-Locked Loops

Feature	Lock Range (Tracking Range)	Capture Range (Acquisition Range)
Definition	The range of frequencies over which a PLL can **maintain** a lock once it is already established.	The range of frequencies over which a PLL can **attain** a lock starting from an unlocked state.
Span	Always **wider** than the capture range ($\Delta\omega_L > \Delta\omega_C$).	Always **narrower** than the lock range.
Dependency	Depends on the **VCO gain** and **Phase Detector range**; independent of the Low-Pass Filter (LPF).	Strictly limited by the **Low-Pass Filter (LPF) bandwidth**.
Analogy	**Holding onto** a moving rope once you have grabbed it.	**Catching** a moving rope while standing still.